Seat no.

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Final Examination of Semester 2 | Year: 2018

Subject: 392152 Chemistry II | Section: 15-18

Date: 22 April 2019 | Time: 10.00 – 12.00

Name:______________________ ID:________________________ Class:________

Directions: The test is designed to measure your comprehension. The test is divided into 2 sections. There will be 8 pages (including this page) and they are worth 50 points.

- Answer the questions on this test papers.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators cannot be used in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

Relative atomic mass of H =1, He =4, C =12, N =14, O =16, F =19, Na =23, S =32, Cl =35.5,
Ar =40, K =39, Cu = 64, Zn = 65.5, Al = 27, P = 30, Fe = 56, I =127

Solvent	Boiling Point (^{o}C)	K_b ($^{o}C/m$)	Freezing Point (^{o}C)	K_f ($^{o}C/m$)
Water	100	0.512	0	1.86
Benzene	80.1	2.53	5.5	5.12
Acetic acid	118.1	3.07	16.6	3.90
Nitroenzene	210.9	5.24	5.7	7.00
Phenol	182	3.56	43	7.40
Camphor	207.4	5.61	178.4	40.0

	Multiple Choice	Part 2.1	Part 2.2	Total
score				

Part 1. Multiple Choice (16 points)

Choose the letter in the answer table that best completes the statement or answers the question.

Question no.	a.	b.	c.	d.
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2				
3				
4				
5				
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Question no.	a.	b.	c.	d.
9				
10				
11				
12				
13				
14				
15				
16				

1. Which statement is correct according to the definition of colligative properties?
   a. Vapor pressure of solution is always less than the pure solvent.
   b. Colligative properties depend on number of solute particles.
   c. Boiling point of solution is always less than the pure solvent.
   d. Molarity is used as the unit of concentration for colligative property calculations.
2. Each of the following solutions is added to equal amounts of water. Which solution will result in the greatest amount of boiling point elevation?
   a. 1.5 mol of $CaCl_2$
   b. 2.5 mol of $KClO_4$
   c. 0 mol of NaCl
   d. 1.25 mol of Na_2SO_4
3. Which statement is correct?
   a. Stock solution is always a high concentration solution.
   b. Colligative properties are the chemical properties of solution.
   c. Dilution is a process that uses for decreasing the concentration of solution.
   d. a. and c. are correct.

4. Which of the following are also examples of colligative properties?
   I. Vapor pressure reduction
   II. Color emission with dissolution of a solute
   III. Osmotic pressure
   IV. Boiling point elevation
   a. I, II
   b. II, III
   c. III, IV
   d. I, III, IV
5. A formula that shows the simplest whole-number ratio of the atoms in a compound is the ________________.
   a. molecular formula.
   b. ideal formula.
   c. structural formula.
   d. empirical formula.
6. Glucose has the molecular formula of $C_6H_{12}O_6$. The correct empirical formula for glucose is ________________.
   a. $C_3H_6O_3$
   b. $C_2H_4O_2$
   c. CH_2O
   d. C_2H_6O
7. Which statement is true?
   a. Formula which gives simplest whole number ratio of atoms is molecular formula.
   b. Empirical formula and molecular formula may be the same in certain cases.
   c. Molecular formula are always different from empirical formula.
   d. all of the above

8. The molecular formula for vitamin C is $C_6H_8O_6$. What is the empirical formula mass?

a. 11　　b. 22　　c. 44　　d. 88

9. Zinc metal reacts with hydrochloric acid by the following reaction:

$Zn + HCl \rightarrow ZnCl_2 + H_2$

how many moles of $ZnCl_2$ are produced if 2 mol of Zn is added to excess HCl 5mol?

a. 0.5 mol　　b. 1.0 mol　　c. 1.5 mol　　d. 2.0 mol

10. Calculate the percent yield for an experiment in which 14.88 g of $SOCl_2$ was obtained in a reaction of 16 g of SO_2 with excess PCl_5. Use the following equation:

$SO_2\,(l) + PCl_5\,(l) \rightarrow SOCl_2\,(l) + POCl_3\,(l)$

a. 50 %　　b. 75 %　　c. 87 %　　d. 90 %

11. Zinc and sulphur react to form zinc sulphide according to the equation.

$Zn + S \rightarrow ZnS$

If 25.0 g of zinc and 30.0 g of sulphur are mixed, which statement is incorrect?

a. Zn is a limiting reagent.

b. S is an excess reactant.

c. 0.38 mol ZnS will be formed.

d. The remaining of an excess reactant is 10.5 g.

12. The actual yield of a reaction is 25 g and the theoretical yield is 40 g. What is the percentage yield?

a. 62.5 %　　b. 75 %　　c. 85 %　　d. 90 %

13. Alkaline earth metals have ____ valence electrons.

a. 1　　b. 2　　c. 4　　d. 8

14. Oxides of most nonmetals combine with water to form __________.

a. an acid　　b. a base　　c. water and a salt　　d. water

15. The outer electron configuration for group 16 is _____________.

a. ns^2　　b. ns^2np^6　　c. ns^2np^1　　d. ns^2np^4

16. Each family or group of elements in a vertical column of the periodic table has elements with chemical characteristics that are _____________.

a. exactly the same　　c. similar.

b. differnt　　d. exactly opposite

Part 2 Write your answer in the space provided for each question (34 points)

1. The freezing point of a glucose solution ($C_6H_{12}O_6$; molar mass= 180.0 g/mole) is - 10.3 ° C What is the molality of the glucose solution? (2 pts)

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2. 171 g of $C_{12}H_{12}O_{11}$ are added to 750 g of H_2O. What is the freezing point and boiling point of this solution? (3 pts)

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3. A solution containing 12.6 g of a nonvolatile and nondissociating solute in 400 g of benzene freezes at 3.6° C . The normal freezing point of benzene is 5.5 ° C. What is the molecular mass of the solute? (3 pts)

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4. Chemical analysis shows the composition of a compound containing carbon and hydrogen, to be 80 % carbon and 20% hydrogen and the molar mass is 30 g. What is its molecular formula? (2 pts)

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5. 0.6 g of a compound containing only C and H gave on complete combustion 1.76 g of CO_2 and 1.08 g of H_2O. Calculate the empirical formula of the compound. (4 pts)

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6. A compound was analyzed and found to contain 13.5 g Ca, 10.8 g O, and 0.675 g H. What is the empirical formula and chemical formula of the compound? If the molar mass of this compound is 73 g/mol. (4 pts)

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7. Determine the limiting reagent if (3 pts)

7.1 1.6 mol Al and 2.9 mol I_2 are used in the reaction : $Al + I_2 \rightarrow AlI_3$

7.2 22 g propane (C_3H_8) reacts with 32 g oxygen in the reaction : $C_3H_8 + O_2 \rightarrow CO_2 + H_2O$

8. Iron reacts with sulfur in the following process: $Fe(s) + S(l) \rightarrow FeS(s)$. If 14 g iron reacts with 16 g sulfur,

8.1 What is the excess reagent?

8.2 How much FeS can theoretically form?

8.3 What is the percentage yield if 18 g of FeS was obtained? (5 pts)

Review of periodic table and oxide/chloride compounds

9. The atomic numbers A and B are 17 and 19, respectively. Write electron configurations for their elements and complete the following table. (4 pts)

A = ..., B = ...

	A		B	
	Oxide	Chloride	Oxide	Chloride
Chemical formula				
Acid-base behavior				

10. Write a balanced chemical equation for the chemical reaction of these oxide compounds reacting with water and define the acid base behavior (4 pts)

Sodium oxide

Chemical equation:

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Carbon dioxide

Chemical equation:

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Monrudee Phongaksorn

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